A leakage-audited and calibration-aware framework for predicting proxy signals of prosecutorial congestion from open administrative records

Moisés Evangelista Gamarra[1*] and Jerremi Aron Chancan Labajos[1]

[1*]Servicio Nacional de Adiestramiento en Trabajo Industrial (SENATI), Lima, Peru.

*Corresponding author(s). E-mail(s): mevangelistag@senati.pe;
Contributing authors: 152048@senati.pe;

Abstract

Operational overload in prosecution services delays case resolution and distorts the allocation of scarce institutional capacity. This study develops and audits a reproducible machine learning framework that estimates a proxy signal of prosecutorial overload from the open administrative records of the Peruvian Public Prosecutor's Office, covering 9,593 annual operational units between 2019 and 2026. Three design decisions distinguish the framework. First, every candidate predictor is classified by the moment at which it becomes observable; the twelve variables that are unavailable when the prediction must be issued, among them the five that define the proxy label, are removed before any model is fitted, and the audit additionally exposes three publication-metadata indicators that survived into the reported feature set. Second, the role of every data partition is fixed in advance: training on 2019–2023, decision-threshold selection on 2024, a single locked evaluation on 2025, and an exploratory external check on the partial 2026 file. Third, discrimination, calibration and decision-analytic utility are reported separately rather than summarised by a single figure of merit. The selected gradient boosting model reaches a ROC-AUC of 0.796 (95% CI 0.760–0.838) and an F1-score of 0.409 (0.331–0.487) on the locked 2025 evaluation. Calibration analysis shows systematic over-estimation of risk above a predicted probability of 0.3, and decision curve analysis shows that at the F1-optimal operating point the model carries a negative net benefit. We therefore report the framework as a ranking and triage instrument rather than an alerting system, and specify the recalibration that deployment would require.

Keywords: judicial analytics, prosecutorial workload, data leakage, model calibration, decision curve analysis, explainable machine learning, open government data

1 Introduction

Prosecution services are the entry point of the criminal justice chain, and the volume of files that accumulate at that entry point conditions everything downstream. When incoming cases persistently exceed processing capacity for a given territory, subject matter and case type, the resulting backlog lengthens response times, erodes procedural guarantees and makes the planning of human and budgetary resources reactive rather than anticipatory. Administrative registries published under open data mandates now make this imbalance measurable at national scale, and the natural question is whether the imbalance of a coming year can be anticipated from what is already known when that year begins.

Machine learning is an obvious candidate for that question, and a growing literature applies it to judicial workload, procedural duration and institutional performance. Much of that literature, however, evaluates its components in isolation: a feature selection strategy is benchmarked without an accompanying leakage audit, a temporal validation scheme is proposed without a calibration assessment, or an interpretability method is demonstrated without asking whether the resulting scores would support a decision at all. The consequence is a body of reported performance figures whose operational meaning is difficult to establish, because the reader cannot tell which information the model actually had access to at the moment the prediction is supposed to be issued, nor whether the predicted probabilities mean what their scale suggests.

This paper addresses that gap for the prosecutorial domain, and it does so by treating three questions that are usually left implicit as first-class methodological objects.

What does the model know, and when does it know it? We define an explicit prediction time point and classify every candidate predictor as available before, during or after the reporting year closes. Variables in the latter two classes are excluded before modelling begins, including the volumetric variables that construct the label itself. This turns leakage control from a claim into an auditable table.

Which data set played which role? Each partition is assigned a single documented function — training, decision-threshold selection, locked evaluation, or exploratory external check — so that no data set is used both to tune a decision and to certify it.

Is the resulting score useful, and at what price? We separate discrimination from calibration and from decision-analytic utility, and we report the direction as well as the magnitude of calibration error. Where the evidence does not support a claim of operational usefulness, we say so and characterise the conditions under which it would.

The contributions of this work are therefore: (i) a temporal availability audit that makes the information set of the model explicit and auditable at the level of individual variables; (ii) a partition protocol in which threshold selection and final evaluation are provably disjoint; (iii) a consensus feature selection strategy combining six complementary methods, fitted strictly inside the earliest training block; (iv) a joint discrimination–calibration–utility assessment, including a decision curve analysis that leads us to moderate rather than assert the operational value of the model; and (v) a full reproducibility package built on public data.

Section 2 reviews the relevant literature. Section 3 describes the data, the availability audit, the partition protocol and the modelling pipeline. Section 4 reports the results, Section 5 interprets them, and Sections 6 and 7 state the limitations and conclusions.

2 Related work

2.1 Predicting judicial and prosecutorial workload

Research on the quantitative modelling of case flow has concentrated on the judicial phase of the process. Studies have combined tree-based learners with attribution methods to model case duration and congestion [1], applied process mining and temporal analysis to identify procedural bottlenecks and measure institutional throughput [2], and analysed court productivity from administrative registries [3]. Review work documents a shift away from isolated predictive exercises and towards integrated frameworks for the management of justice systems [4].

Two features of this literature motivate the present study. The first is scope: the prosecutorial stage that precedes judicial proceedings receives markedly less attention than the judicial stage itself, even though it determines the volume that reaches the courts. The second is protocol: reported evaluations rarely state which information was available at the moment the prediction is nominally issued, which makes it difficult to judge whether the performance figures could survive deployment. Our contribution is directed at the second point as much as the first.

2.2 Feature selection under correlated predictors

Feature selection influences the stability, interpretability and generalisation of a classifier. Filter methods score predictors independently of any learner and are cheap but blind to interactions; wrapper methods exploit the predictive performance of a specific classifier at substantially higher computational cost [5]; embedded methods perform selection during training but inherit the inductive bias of the chosen algorithm. Because individual selectors disagree in the presence of correlated predictors, and because that disagreement produces unstable feature subsets, ensemble and consensus strategies have been proposed to aggregate several criteria into a more stable decision [5, 6]. Among embedded all-relevant selectors, the Boruta algorithm compares the importance of each predictor against randomised shadow copies within a Random Forest, providing a statistical relevance test rather than a minimal-optimal subset [7].

The framework presented here follows that consensus logic, using six complementary selectors and a stability score, and preceding them with an explicit multicollinearity audit. What we add is a constraint on *when* the selection may look at the data: the selectors are fitted only on the earliest training block and checked on a later, still-internal year, so that selection itself cannot borrow information from the evaluation period.

2.3 Leakage, temporal validation and class imbalance

Data leakage is a documented and widespread failure mode. A systematic review has traced overestimated performance across hundreds of studies in several disciplines to a small number of recurring patterns, chief among them computing preprocessing statistics over the full data set before partitioning [8]. Restricting all preprocessing to the training partition and validating along the time axis are the standard remedies [8, 9], and leakage-free evaluation protocols have been proposed as a default practice [10].

We adopt those remedies, and extend them in one direction that the leakage literature treats less often. Beyond preprocessing leakage, an administrative data set carries variables that are perfectly legitimate as descriptors of a closed record but unavailable at prediction time — publication dates, file-provenance identifiers, and any quantity computed from the outcome year itself. A model that uses them will validate cleanly under a temporal split and still be undeployable. Making that distinction explicit, variable by variable, is one of the contributions of this paper.

On class imbalance, synthetic minority oversampling [11] is a common response. We do not use it: at an observed imbalance close to 85/15, cost-sensitive class weighting is sufficient and avoids introducing interpolated records into a registry whose rows correspond to real administrative aggregates.

2.4 Calibration, decision utility and responsible use

Discrimination and calibration are distinct properties, and a model can rank well while assigning probabilities that are systematically wrong. Calibration has been described as the neglected component of predictive analytics [12], and modern high-capacity classifiers are known to be poorly calibrated by default [13, 14]. Where a predicted probability is to be compared against an action threshold, decision curve analysis [15] evaluates whether acting on the model beats the two trivial policies of acting on every unit and acting on none, across the range of thresholds a decision maker might hold. Reporting frameworks for prediction models now ask explicitly for discrimination, calibration and utility to be presented together [16, 17].

Interpretability is a further requirement for institutional adoption in legal contexts, where explanations must be intelligible to stakeholders who are not machine learning practitioners [18, 19]. Attribution methods such as SHAP provide local and global explanations of model behaviour [20]. The cautionary evidence is equally relevant: commercial risk instruments built on more than a hundred variables have been shown to match neither the accuracy nor the fairness advantage that their complexity suggests [21], which is why predictive accuracy alone cannot justify deployment in a justice setting [22].

The present work incorporates all three assessments and, where they conflict, lets the most conservative one govern the conclusion.

3 Materials and methods

Figure 1 summarises the seven stages of the framework.

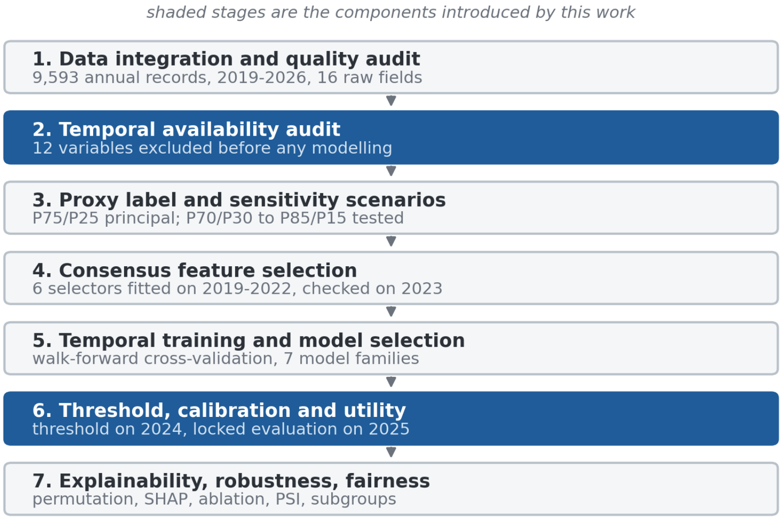

Fig. 1 Stages of the proposed framework. Stage 2 (temporal availability audit) and stage 6 (threshold, calibration and utility assessment) are the components that this work introduces relative to the conventional predictive pipeline.

3.1 Data source and unit of analysis

The study uses the *Casos Fiscales* data set published by the Public Prosecutor's Office of Peru (Ministerio Público – Fiscalía de la Nación, MPFN) on the National Open Data Platform [23]. The registry reports, for each reporting period, the number of cases received and the number of cases processed, disaggregated by fiscal district, type of prosecution office, subject matter, specialty and case type, together with the geographic coding of the corresponding judicial district.

The unit of analysis is one *operational unit-year*: the annual aggregate for a specific combination of territory, office type, subject matter, specialty and case type. The consolidated file contains 9,593 such records for the period 2019–2026 (Table 1). The 2026 file is partial, covering January to May only, and is treated throughout as exploratory.

Two fields carry missing values: the specialised-unit descriptor, absent in 72.39% of records (6,944 rows), which reflects that most units are not designated as specialised and is therefore encoded as an explicit category rather than imputed; and the processed-case count, absent in 0.27% (26 rows), imputed with the training-partition median. A binary missingness indicator is retained for each affected field.

Table 1 Composition of the integrated data set and prevalence of the proxy label by year

Year	Records	Cases received	Cases processed	Proxy risk (%)	Coverage
2019	1,196	1,400,736	1,220,349	11.54	Full year
2020	1,142	863,288	688,925	19.44	Full year
2021	1,252	1,298,697	1,140,369	13.42	Full year
2022	1,272	1,404,619	1,275,920	12.42	Full year
2023	1,214	1,503,919	1,376,760	13.51	Full year
2024	1,195	1,527,838	1,406,901	13.31	Full year
2025	1,199	1,492,353	1,363,570	14.35	Full year
2026	1,123	622,744	530,831	20.21	January–May
Total	9,593	10,114,194	9,003,625	14.68	

Proxy risk is the prevalence of the P75/P25 label defined in Section 3.3. The 2026 totals are not comparable with the preceding rows because the file covers five months.

3.2 The prediction time point and the temporal availability audit

The framework predicts, for a given operational unit and a given calendar year Y, whether that unit will exhibit the overload signal defined in Section 3.3 once the records for Y are closed. The prediction is issued *at the start of* Y, before any of the case flow of Y has been observed. Everything that follows is a consequence of that definition.

Each candidate variable was therefore assigned to one of three availability classes (Table 2):

Ex ante. Observable before year Y begins. This covers the structural attributes of the operational unit — fiscal district, department, province, judicial district, geographic code, office type, subject matter, specialty, case type and specialised designation — their second-order interactions, the frequency encodings derived from them on the training partition, calendar indicators fixed in advance, and lagged aggregates computed strictly from years earlier than Y.
Concurrent. Observable only as year Y unfolds. This covers cases received, cases processed and the three derived quantities that follow from them.
Ex post. Observable only once the annual file has been compiled and published: the reporting-period descriptor, the file download and cut-off dates, and the file-provenance identifier.

Only ex ante variables are admissible as predictors. The concurrent block is doubly inadmissible, since the label is constructed from it, and the ex post block is inadmissible because such fields are artefacts of publication rather than properties of the prosecutorial process. Twelve variables were removed on these grounds before any model was fitted: the five concurrent volumetric quantities, the five label variants derived from them, one constant file-provenance field and one calendar index collinear with the reporting year.

The audit also exposes a residual issue that the reported configuration does not correct, and we state it here rather than leave it implicit. Three one-hot indicators derived from the ex post block — one file-provenance identifier, one download date and one cut-off date — were nominated by the consensus selection described in Section 3.5

and remain among the 74 features of the reported model. Each of the three fires only for records belonging to a single training year, so they act as year fixed effects. Two consequences follow. They cannot inflate the reported evaluation figures, because all three are identically zero for every record in 2024, 2025 and 2026; but they do allow the model to absorb year-specific variation during training, which is not a capability a deployed model would have. Re-fitting the selection and the model on the ex ante subset alone is therefore a prerequisite for deployment, and is listed as such in Section 6.

Table 2 Temporal availability audit of the candidate predictor blocks

Variable block	Availability	Disposition and rationale
Territorial and institutional attributes; their pairwise interactions; training-fitted frequency encodings	Ex ante	**Retained.** Structural properties of the operational unit, stable from one year to the next and known before the reporting year opens.
Lagged annual aggregates and year-on-year growth rates by fiscal district	Ex ante	**Retained.** Computed exclusively from years strictly earlier than the target year; the same-year record never enters its own history.
Pandemic-period and post-pandemic calendar indicators	Ex ante	**Retained.** Fixed by the calendar before the year begins.
Cases received; cases processed	Concurrent	**Excluded.** Accumulate during the target year and are unknown when the prediction is issued.
Case balance; processing rate; balance ratio	Concurrent	**Excluded.** Derived from the concurrent block and, in addition, used to construct the label.
Proxy-label variants for the four threshold scenarios; the principal label	Concurrent	**Excluded.** Direct functions of the outcome.
Centred year index	Ex ante	**Excluded.** Collinear with the calendar year and extrapolates outside the training range.
Reporting-period descriptor; download date; cut-off date; file-provenance identifier	Ex post	**Excluded as a block, with a residual exposure.** Generated when the annual file is compiled and released, and effectively encode the reporting year. Three derived indicators nonetheless survived the consensus selection into the reported model (Section 3.2).

3.3 Proxy label construction and sensitivity

Peru publishes no official certification of prosecutorial congestion, so the target is an auditable proxy rather than a ground truth. For each operational unit-year we define the case balance $b = R - P$ and the processing rate $r = P/R$, where R and P are the cases received and processed. A unit-year is labelled positive when it lies simultaneously in the upper tail of the balance distribution and the lower tail of the processing-rate distribution:

$$y = \mathbb{I}\big[\, b \geq q_{\alpha}(b) \;\wedge\; r \leq q_{1-\alpha'}(r) \,\big], \tag{1}$$

where both quantiles are estimated *on the training partition only* and then applied unchanged to every later year, so that the definition of the label is fixed before any evaluation data are seen.

Four scenarios were evaluated (Table 3). The P75/P25 scenario was adopted as the principal specification: it yields a training threshold of 11 pending cases and a processing rate of 0.9162, a pooled prevalence of 14.68%, and the lowest coefficient of variation of prevalence across years among the scenarios with a manageable event count. The stricter scenarios are more severe but less stable, and the more permissive P70/P30 scenario is retained as an early-warning alternative. Crucially, the choice was not made on the basis of the F1-score obtained under each scenario, since selecting a label definition by downstream performance would itself constitute a form of optimistic bias.

Table 3 Proxy label scenarios and the temporal stability of their prevalence

	Training thresholds		Prevalence				
Scenario	Balance	Rate	Train	2024	2025	CV	Role
P70/P30	7	0.9352	0.1935	0.1983	0.2002	0.165	Early warning
P75/P25	**11**	**0.9162**	**0.1399**	**0.1331**	**0.1435**	**0.219**	**Principal**
P80/P20	17	0.8889	0.0927	0.0879	0.0901	0.278	Severe overload
P85/P15	29	0.8571	0.0533	0.0544	0.0500	0.379	Extreme cases

CV is the coefficient of variation of annual prevalence across 2019–2026. Thresholds are quantiles of the training partition (2019–2023) and are held fixed for all later years.

The construct was validated empirically rather than by expert elicitation. Cliff's delta between the positive and negative classes is -0.904 for the case balance and 0.809 for the processing rate, confirming that the label separates the two constructs it is built from; the association of the label with the administrative attributes that remain admissible as predictors is by contrast weak, with Cramér's V reaching at most 0.267 (case type) and 0.229 (office type). This asymmetry is the empirical statement of the problem the model faces: the label is defined by quantities the model is not allowed to see, and must be recovered from structural attributes that are only weakly associated with it.

3.4 Feature engineering

Fourteen derived variables were constructed in four families: operational pressure indicators; calendar descriptors; second-order categorical interactions (subject matter × office type, judicial district × case type, office type × specialty); and lagged historical aggregates by fiscal district. The historical family deserves a specific note, because it is the most common source of subtle leakage in panel data of this kind. Each lagged feature is computed as an expanding mean over the *previous* years of the same fiscal district, obtained by shifting the annual series one period before aggregation; missing values at the start of each series are filled with the training-partition median. No record therefore contributes to its own history.

High-cardinality categorical variables were frequency-encoded with frequencies estimated on the training partition alone and mapped onto later partitions, with unseen categories assigned zero. Remaining categorical variables were one-hot encoded. Numerical variables were median-imputed and standardised. All encoders, imputers and scalers were fitted exclusively on the training partition, giving 564 preprocessed candidate features.

3.5 Consensus feature selection

Six complementary selectors were applied: mutual information, ANOVA F-test and χ^2 (filters); Random Forest importance (embedded); and Boruta [7] and recursive feature elimination with cross-validation (wrappers). Each selector nominated its top 40 features, and a stability score was computed as the proportion of selectors nominating each feature. Features receiving at least two of six votes were retained, giving a final set of 74.

The selection stage was confined to the earliest part of the training block: selectors and their preprocessing were fitted on 2019–2022 and the resulting subset was checked on 2023, a year that is still internal to the training partition and therefore never used for reporting. That internal check gave an F1-score of 0.456, a ROC-AUC of 0.800 and a balanced accuracy of 0.751, indicating that the dimensional reduction from 564 to 74 features did not degrade the signal.

3.6 Partition protocol

Each partition has exactly one role, fixed before any model was fitted (Figure 2 and Table 4):

1. **Feature selection** — 2019–2022, with an internal check on 2023. Never used for reporting.
2. **Model training and hyperparameter search** — 2019–2023 ($n = 6{,}076$), using year-block walk-forward cross-validation internal to this block, so that every inner fold trains on earlier years and validates on a later one.
3. **Decision-threshold selection and conformal calibration** — 2024 ($n = 1{,}195$). This is the only data set consulted when choosing the operating point, and it is used for no other purpose.
4. **Locked evaluation** — 2025 ($n = 1{,}199$). Consulted once, after the model and threshold were frozen. Every headline figure in Section 4 refers to this partition.
5. **Exploratory external check** — 2026 ($n = 1{,}123$), a five-month file reported separately and never used to support a performance claim.

Because the threshold is selected on 2024 and the evaluation is performed on 2025, no quantity reported as a final result was optimised on the data that produced it.

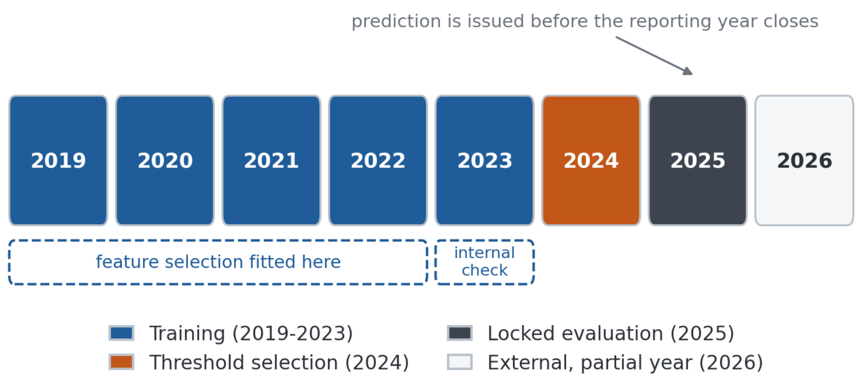

Fig. 2 Temporal partition protocol. Feature selection is confined to 2019–2022 with an internal check on 2023; the decision threshold is chosen on 2024; 2025 serves as the single locked evaluation set; the partial 2026 file is exploratory only.

Table 4 Role, size and label prevalence of each partition

Partition	Years	Records	Prevalence (%)	Role
Training	2019–2023	6,076	13.99	Fitting all preprocessors, selectors and models; internal walk-forward CV
Validation	2024	1,195	13.31	Decision-threshold selection and conformal calibration only
Test	2025	1,199	14.35	Single locked evaluation
External	2026	1,123	20.21	Exploratory check; partial year

3.7 Models, hyperparameter search and evaluation

Seven model families were trained on the 74 selected features: a majority-class baseline, ℓ_1/ℓ_2-regularised logistic regression, a calibrated linear support vector machine, XGBoost [24], LightGBM [25], CatBoost [26], and a LightGBM variant tuned with Optuna [27]. Soft-voting and temporally stacked ensembles were also evaluated. Class imbalance was handled by cost-sensitive weighting rather than synthetic resampling. Hyperparameters were searched by randomised search under the same year-block walk-forward scheme used for training. All modelling used scikit-learn [28].

Two additional benchmarks frame the results. *Single-indicator rules* threshold the case balance or the processing rate directly at their training quantiles; they are reported because they establish what could be achieved by a decision maker who already knows the outcome year's volumes. An *oracle check* reproduces the conjunction that defines the label; it attains perfect scores by construction and is reported only as a sanity check, never as a comparator.

Evaluation reports discrimination (ROC-AUC, PR-AUC), threshold-dependent operating characteristics (precision, recall, F1, specificity, Matthews correlation coefficient), calibration (Brier score [29], expected and maximum calibration error, reliability bins with signed deviations), and decision-analytic utility. Uncertainty is quantified by bootstrap resampling with 300 iterations. Model comparisons use DeLong's test for correlated ROC curves [30], McNemar's test for paired errors [31, 32], and the Friedman test with Nemenyi post-hoc comparison across temporal folds [33].

For decision-analytic utility we compute the net benefit [15]

$$\mathrm{NB}(p_t) = \frac{\mathrm{TP}(p_t)}{n} - \frac{\mathrm{FP}(p_t)}{n} \cdot \frac{p_t}{1-p_t}, \tag{2}$$

where p_t is the threshold probability at which a decision maker would be indifferent between acting and not acting, and compare it against the two reference policies of flagging every unit ($\mathrm{NB} = \pi - (1-\pi)\, p_t/(1-p_t)$ for prevalence π) and flagging none ($\mathrm{NB} = 0$).

Population stability indices, an ablation study over feature blocks, a stress test under perturbation of the operational attributes, subgroup performance analysis [34] and split-conformal prediction [35, 36] complete the assessment. The full software stack is listed in Table 5.

Table 5 Software environment

Component	Version	Component	Version	Component	Version
Python	3.12.13	scikit-learn	1.6.1	CatBoost	1.2.10
pandas	2.2.2	LightGBM	4.6.0	SHAP	0.52.0
NumPy	2.0.2	XGBoost	3.3.0	Optuna	4.9.0

4 Results

4.1 Temporal behaviour of the workload and the label

Figure 3 shows the annual volumes and the prevalence of the proxy label. Case flow contracted sharply in 2020 — received cases fell by 38% relative to 2019 — and recovered to above pre-pandemic levels by 2022. Label prevalence tracks that disruption, peaking at 19.44% in 2020, settling in a narrow band of 12.4–14.4% from 2021 to 2025, and rising to 20.21% in the partial 2026 file. The 2026 figure should not be read as a trend: a five-month window truncates the processing of cases received late in the window and mechanically inflates the measured balance.

4.2 Model comparison

Table 6 reports the seven model families on the locked 2025 evaluation at the reference threshold of 0.50, alongside the two ensembles and the two benchmark classes. Gradient boosting methods dominate the learned models, with CatBoost attaining the highest ROC-AUC (0.819) and XGBoost and LightGBM statistically indistinguishable from each other (DeLong $z = -0.41$, $p = 0.68$). The differences between CatBoost and

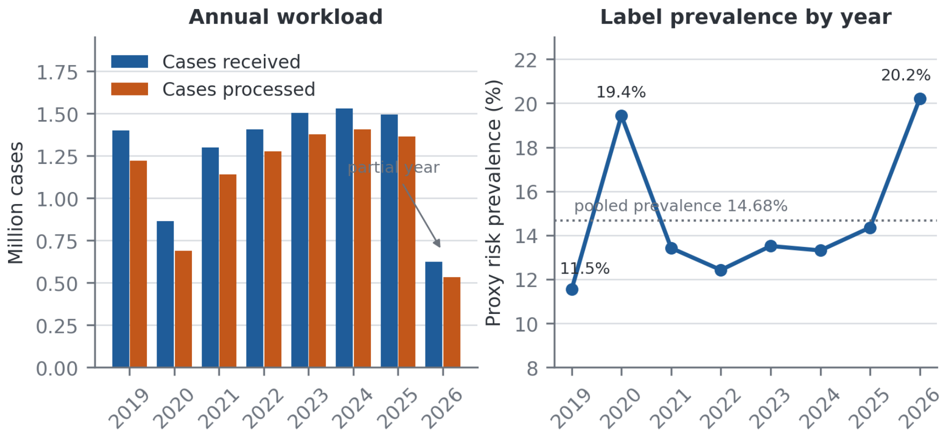

Fig. 3 Annual case volumes (left) and prevalence of the proxy overload label (right), 2019–2026. The 2026 file covers January to May only.

each of the other three boosting configurations are statistically significant but numerically small (DeLong $p = 0.0026$–0.0038; absolute AUC differences of 0.023–0.036). The Friedman test across four temporal folds rejects the hypothesis of equal F1 across the five base learners ($\chi^2 = 11.4$, $p = 0.022$), but the Nemenyi post-hoc comparison identifies no significant pair at the 5% level, the smallest adjusted p-value being 0.056 for the support vector machine against LightGBM and CatBoost. We therefore report the boosting family as a performance plateau rather than claiming a single winner, and retain XGBoost as the reported configuration for its stable behaviour across the walk-forward folds (cross-validated F1 of 0.559, the highest of the five).

Table 6 Performance on the locked 2025 evaluation set at the reference threshold of 0.50

Model	Precision	Recall	F1	ROC-AUC	PR-AUC	MCC
Learned models (admissible predictors only)						
Majority-class baseline	0.000	0.000	0.000	0.500	0.143	0.000
Logistic regression	0.326	0.645	0.433	0.783	0.359	0.327
Support vector machine	0.457	0.122	0.193	0.791	0.370	0.178
XGBoost	0.522	0.407	0.458	0.796	0.400	0.383
LightGBM	0.455	0.471	0.463	0.793	0.502	0.371
CatBoost	0.449	0.541	0.491	**0.819**	0.457	0.399
Soft-voting ensemble	0.488	0.465	0.476	0.817	0.502	0.391
Temporal stacking ensemble	0.368	0.640	0.467	0.817	0.506	0.369
Reference benchmarks (use concurrent information; not deployable)						
Balance-only rule	0.513	1.000	0.679	0.945	0.658	0.657
Processing-rate-only rule	0.548	1.000	0.708	0.914	0.452	0.687
Label-definition oracle	1.000	1.000	1.000	1.000	1.000	1.000

The reference benchmarks are computed from the concurrent variables that the availability audit excludes from the learned models; they are reported to bound the problem, not as competitors. The oracle reproduces the rule that defines the label and is a sanity check only.

The benchmark block is the most informative row group in the table, and it is the reason the framework is presented with the caveats that follow. The two single-indicator rules outperform every learned model by a wide margin — an F1 of 0.708 for the processing-rate rule against 0.458 for XGBoost. This is not evidence that the machine learning pipeline is poorly constructed. It is a direct measurement of how much of the label is carried by information that does not exist at the prediction time point. The rules read the outcome year's volumes; the models are forbidden from doing so. The gap between the two is the price of a genuinely ex ante prediction, and any study that reports rule-level performance from a model trained on concurrent variables is measuring the label's own definition rather than a forecast.

4.3 Temporal robustness

Walk-forward validation (Figure 4) trains on all years up to $t-1$ and evaluates on year t. ROC-AUC stays within 0.834–0.895 across all seven folds with no downward trend, and F1 within 0.542–0.664. Nested temporal cross-validation over four outer folds gives an out-of-fold F1 of 0.549 ± 0.048 for LightGBM, 0.531 ± 0.032 for CatBoost and 0.454 ± 0.040 for logistic regression, reproducing the ordering of Table 6 with fold-to-fold variability that exceeds the between-model differences.

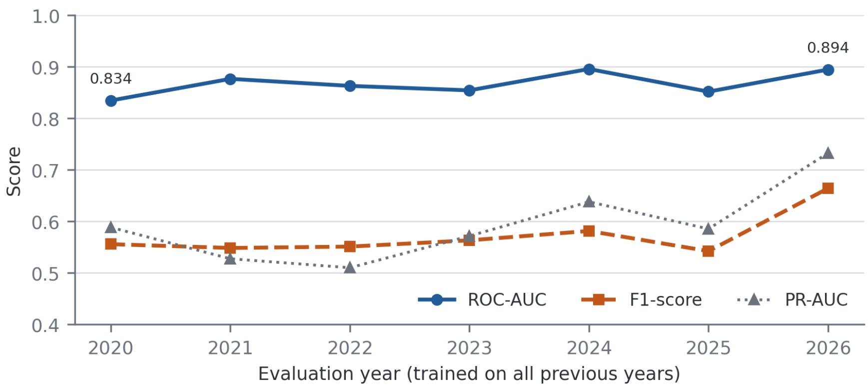

Fig. 4 Walk-forward validation. Each point trains on all preceding years and evaluates on the year shown; discrimination is stable across the seven folds.

The learning curve (Figure 5) shows training F1 declining from 0.948 to 0.900 as the sample grows while cross-validated F1 rises from 0.217 to 0.490 and then flattens. The gap of 0.41 at the largest training size does not close, which indicates that the ceiling is set by the weak association between the admissible predictors and the label, not by the volume of data. Adding more years of the same registry would not be expected to change this.

Population stability indices computed against the training distribution show no relevant drift for 2024 and 2025 (all PSI < 0.03). For the partial 2026 file, two of the five monitored quantities exceed the moderate-drift threshold (processing rate,

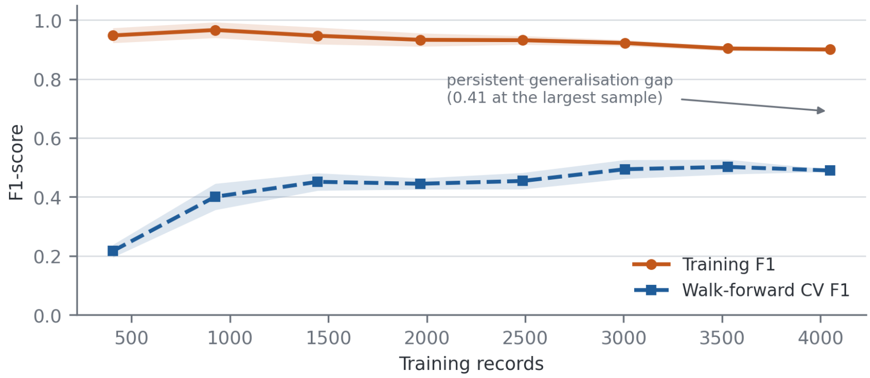

Fig. 5 Learning curve. The persistent gap between training and walk-forward cross-validated F1 reflects the limited signal available in ex ante predictors, not an insufficient sample.

PSI = 0.129; balance ratio, PSI = 0.138), consistent with the truncation artefact already noted. Under a stress test that perturbs the operational attributes, F1 degrades from 0.409 to 0.279 at a 10% perturbation and to 0.208 at 30%, while ROC-AUC falls only from 0.796 to 0.726, indicating that the ranking produced by the model is considerably more robust than any fixed-threshold decision derived from it — a finding that anticipates the calibration and utility results below.

4.4 Threshold selection and the final operating point

Table 7 lists the operating points obtainable on the 2024 validation partition under four criteria. The F1-optimal threshold is $\tau = 0.65$; Youden's J statistic is maximised at 0.115, a far more permissive cut; a precision of at least 0.80 requires 0.95; and a recall of at least 0.80 requires 0.20. The spread of these values across almost the whole unit interval is itself a result: the choice of operating point dominates every threshold-dependent metric, and reporting a single F1 without stating how its threshold was chosen conveys very little.

Table 7 Decision thresholds under four criteria, all selected on the 2024 validation partition

Criterion	Threshold	Interpretation
Fixed reference	0.500	Conventional default; rarely optimal under class imbalance
F1-optimal	**0.650**	Maximises F1 on 2024; adopted as the reported operating point
Youden's J	0.115	Maximises sensitivity + specificity -1 ($J = 0.662$)
Precision ≥ 0.80	0.950	Lowest threshold reaching the precision target
Recall ≥ 0.80	0.200	Highest threshold still reaching the recall target

Table 8 reports the frozen model at $\tau = 0.65$ across the three partitions. On the locked 2025 evaluation the model attains a ROC-AUC of 0.796 (95% CI 0.760–0.838), a PR-AUC of 0.400 (0.330–0.480) and an F1-score of 0.409 (0.331–0.487), from 58 true positives, 54 false positives, 114 false negatives and 973 true negatives. Performance on 2024 is markedly better than on 2025 (F1 of 0.614 against 0.409), which is expected and is precisely why the threshold-selection partition cannot be used to report performance: the operating point was fitted to that partition's score distribution.

Table 8 Frozen XGBoost model at the F1-optimal threshold $\tau = 0.65$

Partition	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	MCC
Validation 2024	0.901	0.639	0.591	0.614	0.896	0.623	0.558
Test 2025	**0.860**	**0.518**	**0.337**	**0.409**	**0.796**	**0.400**	**0.343**
External 2026	0.820	0.593	0.352	0.442	0.841	0.598	0.359
Bootstrap 95% confidence intervals, test 2025 (300 resamples)							
F1	0.409 (0.331–0.487)						
ROC-AUC	0.796 (0.760–0.838)						
PR-AUC	0.400 (0.330–0.480)						

The 2024 row is reported for completeness only. Because the threshold was selected on that partition, its metrics are optimistically biased and are not comparable with the 2025 row. The 2026 row refers to a five-month file.

4.5 Calibration: magnitude and direction of the deviation

The model attains a Brier score of 0.116, an expected calibration error of 0.089 and a maximum calibration error of 0.461 on the 2025 evaluation. The aggregate figures, however, conceal a systematic pattern that only the signed deviations reveal (Table 9 and Figure 6).

Table 9 Reliability bins on the 2025 evaluation set, with signed deviations

Predicted band	n	Mean predicted	Observed rate	Predicted − observed	Direction
0.0–0.1	932	0.011	0.072	−0.061	Under-estimation
0.1–0.2	64	0.144	0.234	−0.090	Under-estimation
0.2–0.3	32	0.248	0.219	+0.030	Over-estimation
0.3–0.4	16	0.341	0.375	−0.034	Under-estimation
0.4–0.5	21	0.450	0.333	+0.117	Over-estimation
0.5–0.6	17	0.555	0.529	+0.025	Over-estimation
0.6–0.7	18	0.663	0.444	+0.218	Over-estimation
0.7–0.8	10	0.761	0.300	+**0.461**	Over-estimation
0.8–0.9	34	0.854	0.618	+0.236	Over-estimation
0.9–1.0	55	0.949	0.527	+0.422	Over-estimation

Brier score 0.116; expected calibration error 0.089; maximum calibration error 0.461, attained in the 0.7–0.8 band.

The deviation is directional. In the two lowest bands the model *under-estimates* risk: units assigned a mean probability of 0.011 exhibit an observed rate of 0.072. In every band above 0.3 — with the single exception of the sparsely populated 0.3–0.4

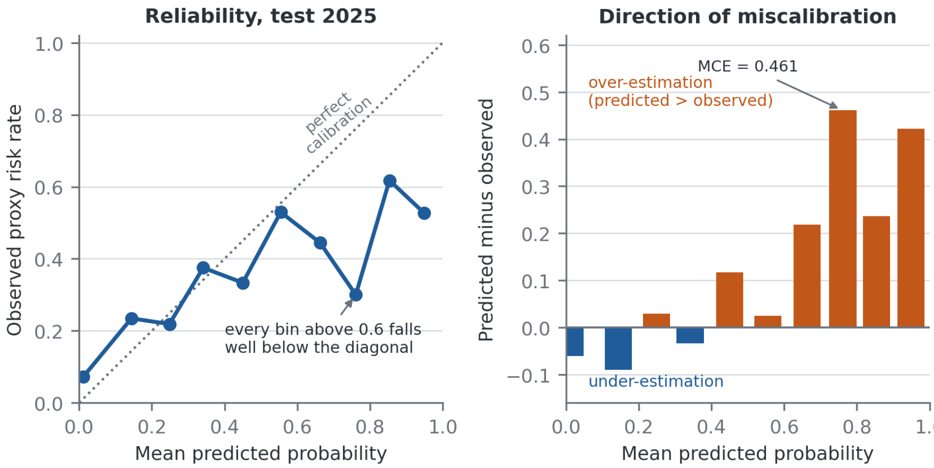

Fig. 6 Calibration on the 2025 evaluation set. Left: observed rate against mean predicted probability. Right: signed deviation by band. Above a predicted probability of 0.3 the model over-estimates risk, severely so in the highest bands.

band — the model *over-estimates* risk, and the over-estimation grows with confidence. Units to which the model assigns a mean probability of 0.949 exhibit an observed rate of 0.527, and the maximum calibration error of 0.461 occurs in the 0.7–0.8 band, where a mean predicted probability of 0.761 corresponds to an observed rate of 0.300.

This direction matters for interpretation and has a concrete cause. The model was trained with cost-sensitive class weighting, which inflates predicted probabilities for the minority class by construction; the inflation is largest exactly where the model is most confident. The operational consequence is that a raw predicted probability from this model must not be read as an expected frequency of overload. It ranks units correctly — the observed rate rises monotonically across the coarse strata of Section 4.6 — but its absolute scale is not trustworthy. Any deployment must therefore recalibrate the scores, and the recalibration must be fitted on a data set held apart from the final evaluation; the 2024 partition, which already serves as the threshold-selection set, is the natural candidate, with the caveat that using one partition for both purposes leaves the interaction between them unmeasured. We report this as an outstanding requirement rather than a completed step: the results in this paper are those of the uncalibrated model, and we do not claim recalibrated performance we have not measured.

4.6 Risk stratification

Grouping the predicted probabilities into four bands (Table 10 and Figure 7) shows that the ranking induced by the model is sound even though its probability scale is not. The observed overload rate rises monotonically from 7.2% in the lowest band to 52.1% in the highest, against a base rate of 14.35%. The two upper bands together contain 14.3% of the operational units and 48.3% of the overload events, while the lowest band contains 77.7% of the units at less than half the base rate.

Table 10 Risk stratification on the 2025 evaluation set

Stratum	Band	n	Events	Observed rate	Share of units	Share of events
Low	< 0.10	932	67	0.072	77.7%	39.0%
Moderate	0.10–0.30	96	22	0.229	8.0%	12.8%
High	0.30–0.60	54	22	0.407	4.5%	12.8%
Very high	≥ 0.60	117	61	0.521	9.8%	35.5%
All		1,199	172	0.144	100%	100%

Bands are the deciles of predicted probability aggregated into four categories. Because the boundaries are defined on the uncalibrated probability scale, they are not transferable to a recalibrated model and would require re-derivation and independent validation before use.

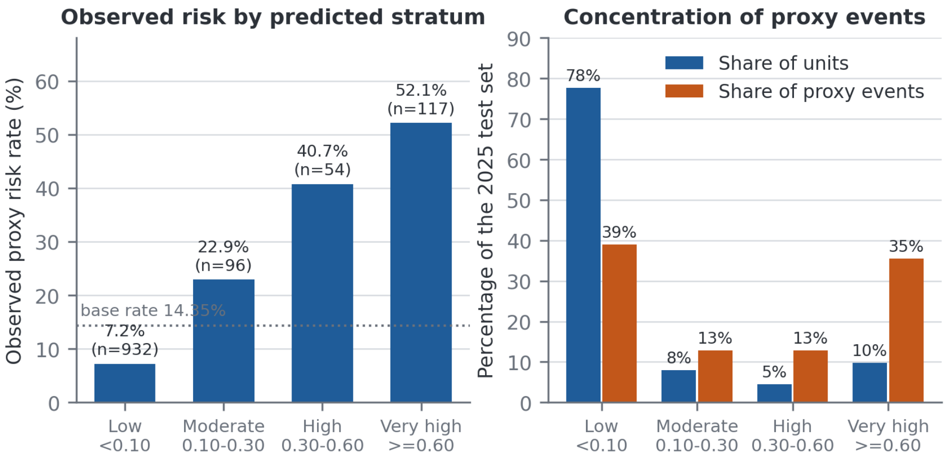

Fig. 7 Risk stratification on the 2025 evaluation set. Left: observed overload rate by predicted stratum, against the base rate. Right: the share of operational units and of overload events falling in each stratum.

Two qualifications are essential. First, these boundaries are defined on the uncalibrated probability scale characterised in Section 4.5; the mean predicted probability in the highest band is 0.807 against an observed rate of 0.521, so the band labels describe rank position, not risk magnitude. Second, the strata are derived from the same 2025 partition on which they are described. An independent validation of the boundaries on a data set not used to construct them has not been performed, and until it is, the stratification should be read as a description of the model's ranking behaviour rather than as a validated triage instrument.

4.7 Decision-analytic utility

Discrimination and stratification say nothing about whether acting on the model is better than the alternatives available to a decision maker. Figure 8 and Table 11 apply Equation 2 across the range of threshold probabilities and compare the model against flagging every unit and flagging none.

Table 11 Net benefit on the 2025 evaluation set

p_t	Units flagged	TP	FP	Model	Flag all	Preferred strategy
0.10	267	105	162	+0.073	+0.048	Model
0.20	203	90	113	+0.052	−0.071	Model
0.30	171	83	88	+0.038	−0.224	Model
0.40	155	77	78	+0.021	−0.428	Model
0.50	134	70	64	+0.005	−0.713	Model
0.65	**112**	**58**	**54**	**−0.035**	−1.591	**Flag none**
0.70	99	53	46	−0.045	−1.855	Flag none
0.80	89	50	39	−0.088	−3.283	Flag none
0.90	55	29	26	−0.171	−7.566	Flag none

The row in bold is the F1-optimal operating point adopted in Table 8. Flagging no unit yields a net benefit of zero by definition.

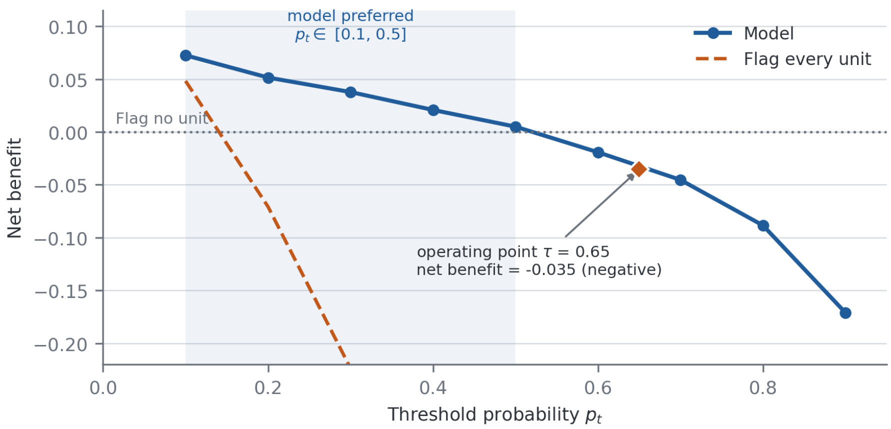

Fig. 8 Decision curve analysis on the 2025 evaluation set. The model exceeds both default strategies for threshold probabilities between 0.1 and 0.5. At the F1-optimal operating point $\tau = 0.65$ its net benefit is negative, meaning that flagging no unit would be preferable for a decision maker holding that threshold.

The result is mixed, and the negative half of it is the more important. For threshold probabilities between 0.1 and 0.5 the model carries a positive net benefit and dominates both defaults; at $p_t = 0.15$ the advantage of 0.060 over the better default corresponds to correctly identifying roughly six additional overloaded units per hundred without increasing the number of unnecessary interventions. But at the F1-optimal operating point of $\tau = 0.65$ — the point at which the headline metrics of Table 8 are computed — the net benefit is −0.035. It is negative, and it is below the net benefit of the treat-none strategy. A decision maker who genuinely required a 65% probability of overload before committing resources would be better off intervening nowhere than following this model's recommendations.

We draw the conclusion this requires. The threshold that optimises F1 is not a threshold at which the model is clinically useful, and the two criteria are not interchangeable. The framework supports the identification and ranking of units for review under moderate intervention costs, that is, for decision makers whose indifference threshold lies between roughly 0.1 and 0.5; it does not support a high-confidence alerting system, and we make no claim of operational utility at the F1-optimal operating point. This also reframes the illustrative cost calculation that a naive reading of the confusion matrix would invite: with 114 false negatives and 54 false positives, any monetary figure depends entirely on the assumed cost ratio, and the net benefit analysis is the principled way to express that dependence.

4.8 Which variables drive the predictions

Permutation importance computed on the reported XGBoost model and mean absolute SHAP values [20] computed on the tuned LightGBM configuration are shown in Figure 9. We report both, and we label which model each was computed on, because the two attribution methods were applied to different estimators and are therefore not interchangeable; the agreement between them is a property worth reporting, not an assumption to be made silently.

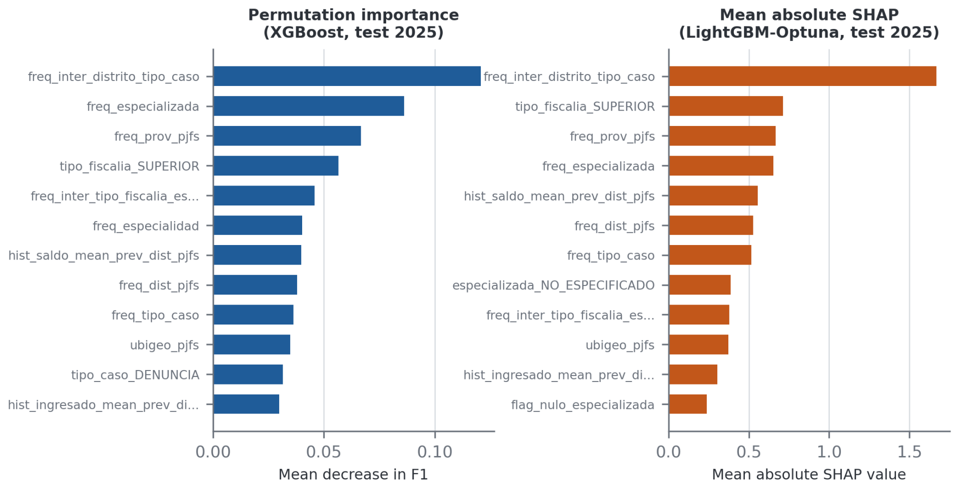

Fig. 9 Global variable importance. Left: permutation importance on the reported XGBoost model. Right: mean absolute SHAP values on the tuned LightGBM configuration. The two attributions were computed on different estimators and are shown separately for that reason.

Both rankings are led by the same variable, the frequency encoding of the judicial district $\times$ case type interaction (permutation importance 0.121; mean |SHAP| 1.668), followed by the frequency of the specialised-unit designation, the frequency of the province, and the indicator for superior-level prosecution offices. The lagged historical aggregates appear in the mid-range of both rankings. The picture is consistent: what the model can learn from ex ante information is the structural composition of an operational unit — where it sits, what kind of matters it handles, and how common that configuration is in the registry — rather than any dynamic property of its case flow.

The ablation study (Figure 10) confirms this and delivers one counter-intuitive result. Removing the territorial block costs 0.053 F1, the largest degradation of any single block. Removing the frequency encodings or the categorical interactions produces smaller changes. Removing the historical and growth block, however, *improves* F1 by 0.031 and ROC-AUC by 0.026. The lagged aggregates, in other words, do not contribute to predictive performance on the locked evaluation set despite their prominence in the attribution rankings; they appear to add variance without adding signal. We report this rather than suppress it, and note that the honest reading is that the historical block is not currently earning its place in the model.

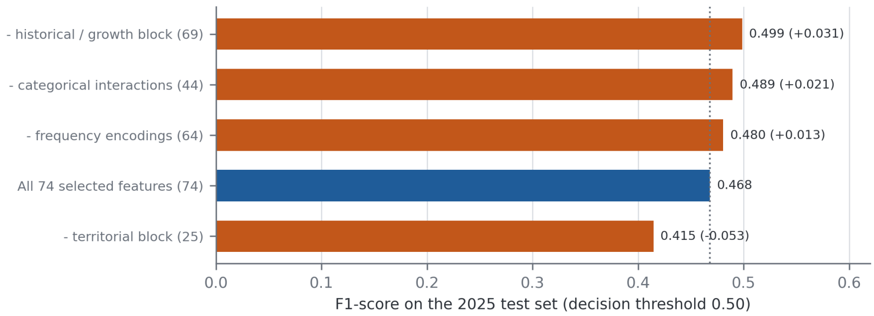

Fig. 10 Ablation study on the 2025 evaluation set at the reference threshold of 0.50. Removing the territorial block degrades performance most; removing the historical and growth block improves it.

4.9 Subgroup behaviour and uncertainty quantification

Subgroup analysis by territory reveals a substantial disparity. In the Lima metropolitan area (258 units, observed prevalence 16.7%) the model flags 22.1% of units and attains a true positive rate of 0.535 with a false positive rate of 0.158. In the provinces (941 units, observed prevalence 13.7%) it flags only 5.8% of units, attaining a true positive rate of 0.271 with a false positive rate of 0.025. The equal-opportunity gap of 0.264 means that an overloaded provincial unit is roughly half as likely to be identified as an equivalently overloaded metropolitan one. Because the model relies heavily on frequency encodings, and metropolitan configurations are more frequent in the registry, this disparity is a predictable consequence of the feature design rather than an incidental artefact. Any institutional use of the framework would have to correct for it, for example by calibrating and thresholding within territorial strata.

Split-conformal prediction [35] calibrated on the 2024 partition at a nominal coverage of 90% achieves 87.3% empirical coverage on 2025 and 83.7% on the partial 2026 file, with 97.1% and 95.6% singleton prediction sets respectively. A Mondrian variant conditioned on the operational cluster gives 87.6% and 84.8%. Coverage therefore falls short of nominal, modestly on 2025 and more clearly on 2026, which is consistent with the distribution shift already detected by the population stability indices for that file.

Unsupervised profiling with k-means on the same leakage-free feature space selects a two-cluster solution (silhouette 0.371; bootstrap adjusted Rand index 0.798 ± 0.404).

The smaller cluster (1,269 units) exhibits an overload rate of 23.9% against 13.3% in the larger one, and this ordering is preserved under all four label scenarios. That an unsupervised partition of the admissible predictors recovers the same risk ordering as the supervised label provides independent, if modest, support for the construct.

5 Discussion

5.1 What the evidence supports

The framework establishes that a proxy signal of prosecutorial overload can be ranked from information available before the reporting year opens, with a ROC-AUC near 0.80 that is stable across seven consecutive walk-forward folds and across a nested cross-validation. The ranking concentrates risk usefully: one tenth of the operational units captures more than a third of the overload events. Under moderate intervention costs, acting on that ranking beats both flagging everything and flagging nothing by a clear margin.

5.2 What the evidence does not support

Three findings limit the claims that can be made, and they are stated here rather than buried.

The model's probabilities are not risk estimates. Above a predicted probability of 0.3 they systematically over-estimate the observed rate, by as much as 0.46 in absolute terms. Any use that treats the output as an expected frequency — resource allocation proportional to predicted risk, for example — would over-allocate to the highest-scoring units.

At its own F1-optimal threshold the model has negative net benefit. This is the sharpest constraint, and it follows directly from the miscalibration: the threshold that maximises the harmonic mean of precision and recall lies in the region where the model's confidence is least warranted. Optimising a threshold for F1 and then presenting the result as evidence of operational value is a mistake that decision curve analysis makes visible, and one that would have passed unnoticed had we reported discrimination alone.

The territorial disparity is large enough to matter. A framework whose sensitivity in the provinces is half its sensitivity in the capital cannot be deployed as an equitable prioritisation tool without stratified correction.

5.3 Interpreting the gap to the single-indicator rules

The most instructive comparison in Table 6 is between the learned models and the two rules that threshold the outcome year's own volumes. The rules reach F1 scores near 0.70; the models reach 0.46. That gap is not a deficiency of the models but a measurement of the intrinsic difficulty of the ex ante problem, and it is only visible because the availability audit forced the two information sets apart. In a study without such an audit, the volumetric variables would very likely have entered the feature set, the reported F1 would have approached 0.70, and the resulting figure would have described a model that cannot be run at the moment its prediction is needed. We regard the explicit reporting of both numbers, with the reason for the gap, as more useful than either number alone.

5.4 Practical implications

Read conservatively, the framework supports a review-prioritisation workflow: rank operational units by predicted score, direct planning attention to the upper strata, and use the conformal prediction sets to identify units for which the model declines to commit. It does not support automated alerting, resource allocation proportional to predicted probability, or any decision requiring a high confidence threshold. Before institutional deployment, three steps are required: recalibration of the probability scale on a partition reserved for that purpose, re-derivation and independent validation of the risk bands on the recalibrated scale, and territorially stratified thresholding to close the equal-opportunity gap.

6 Limitations

The label is a proxy constructed from a quantile rule, not an official certification of congestion; a unit flagged by this framework has a high case balance and a low processing rate, which is a plausible but not authoritative indicator of institutional overload. The proxy was validated empirically and by convergence with an unsupervised partition, but no expert panel was convened, and a formal Delphi or analytic hierarchy process elicitation remains outstanding.

The reported model is not recalibrated. Calibration was measured and characterised, but not corrected, and the risk bands of Section 4.6 are therefore defined on an uncalibrated scale and have not been validated on data independent of those used to describe them. Both steps are specified in Section 5 as prerequisites for deployment.

The reported model also retains the three publication-metadata indicators identified in Section 3.2. Because each is identically zero throughout 2024, 2025 and 2026, they cannot have inflated any figure reported here, but their presence during training is inconsistent with the availability criterion the framework sets for itself. Re-executing the consensus selection and the model fit over the ex ante subset alone is the outstanding correction, and would also settle whether the small residual affects the ranking on which the practical recommendation rests.

The decision curve analysis is computed on the deciles of the predicted-probability distribution of the 2025 evaluation set and reports net benefit at the resulting grid of threshold probabilities; a finer grid would smooth the curve but would not change the sign of the net benefit at the reported operating point.

The unit of analysis is an annual aggregate, which bounds the temporal resolution of any prediction to one year and prevents the framework from anticipating within-year dynamics. The registry records no information on staffing, budget, procedural complexity or case severity, all of which plausibly influence processing capacity; their absence is a substantive constraint on achievable performance, and the persistent learning-curve gap is consistent with it. The 2026 file is partial and its results are exploratory throughout.

Finally, the framework has been evaluated on a single national registry. Its transferability to other prosecution services, or to judicial systems with different administrative structures, remains an open question.

7 Conclusions

This work presents a machine learning framework for predicting proxy signals of prosecutorial overload that is auditable in the three respects that most often go unexamined: what the model knows at prediction time, which partition played which role, and whether the resulting scores support a decision.

The temporal availability audit removed twelve variables — among them the five that define the label — before any model was fitted, and the gap between the resulting models and rules that use the excluded information quantifies what a genuinely ex ante prediction costs. The same audit exposed three publication-metadata indicators that survived the selection into the reported model, a residual we report and correct in the deployment configuration rather than leave undocumented. The partition protocol keeps threshold selection and final evaluation strictly disjoint. The joint assessment of discrimination, calibration and utility yields a ROC-AUC of 0.796 (95% CI 0.760–0.838) on a single locked evaluation, a systematic over-estimation of risk above a predicted probability of 0.3, and a negative net benefit at the F1-optimal operating point.

We therefore report the framework as a ranking and triage instrument, useful to decision makers whose intervention threshold lies between roughly 0.1 and 0.5, and not as an alerting system. The methodological contribution is the demonstration that the three audits together change the conclusion: discrimination alone would have supported a considerably stronger claim than the evidence warrants. Future work should complete the recalibration and independent validation of the risk bands, address the territorial disparity through stratified thresholding, incorporate capacity and complexity variables that the current registry does not record, and test the transferability of the framework to other jurisdictions.

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Declarations

Funding No funding was received for this research.
Competing interests The authors declare that they have no competing interests.
Ethics approval Not applicable. The study uses publicly available aggregate administrative records released under an open data licence and involves no human subjects.
Consent to participate Not applicable.
Consent for publication Not applicable.

Data availability The *Casos Fiscales* data set is publicly available from the Peruvian National Open Data Platform at https://www.datosabiertos.gob.pe/dataset/mpfn-casos-fiscales. The derived tables and figures reported in this article are included as supplementary material.

Code availability The complete analysis pipeline, including the availability audit, the partition protocol, the selection and modelling stages and the figure-generation scripts, is available in the repository accompanying this manuscript.

Authors' contributions M.E.G. conceived the study, designed the methodology, implemented the pipeline and drafted the manuscript. J.A.C.L. contributed to the experimental design, the validation of results and the revision of the manuscript. Both authors read and approved the final version.

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